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Event-Driven Intrusion Detection Systems using Spiking Neural Networks for Edge and IoT Security

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***Abstract*— The rapid growth of Internet of Things (IoT) devices and the increasing difficulty of cyber-attacks highlight the need for novel, energy-efficient, and real-time Intrusion Detection Systems (IDS) capable of running on resource constrained edge devices. In this work, we present a Spiking Neural Networks (SNN) -based event-driven intrusion detection framework inspired by neuromorphic computing principles. Our event-driven detection framework is unique compared to traditional continuous data-based IDS because it uses sparse, temporal event-based encoding to detect malicious activity with high precision and low power. We implement a biologically plausible, unsupervised SNN architecture that learns the spatio-temporal patterns of both normal and anomalous behavior using Spike-Timing Dependent Plasticity (STDP) under the concept of continuous learning to keep up with the changing threat landscape. In addition, we provide an innovative technique for spike-based feature encoding that preserves critical temporal elements while relieving some communication overhead. This work presents the first step towards implementing a reliable, low-latency, and energy-efficient neuromorphic security solution at the edge that integrates brain-inspired AI and IoT security applications. We evaluate our proposed framework on benchmark IoT intrusion datasets using a customized neuromorphic simulator that is designed to be deployed at the edge. Our results illustrate that this system is capable of achieving superior detection accuracy (as high as 97.8%), while consuming up to 55% less energy than traditional deep learning-based IDS. Furthermore, the SNN's event-based processing lends itself to real-time detection with latency on the order of milliseconds, making it ideal for low-latency in Cyber-Physical Systems (CPS) and smart edge ecosystems.**

***Keywords*— Cyber-Physical Systems (CPS), Spike-Timing Dependent Plasticity (STDP), Cyber-Physical Systems (CPS), Intrusion Detection Systems (IDS) and Internet of Things (IoT)**

I. INTRODUCTION

The very fast growth of the Internet of Things (IoT) has resulted in an era of pervasive computing, with billions of connected devices generating huge amounts of data in real-time. These devices, which typically operate in edge networks with limited resources, are the cornerstones of today's smart infrastructure, ranging from industrial automation and cyber-physical systems (CPS) to smart homes and health systems [1-2]. However, because of their extensive connectivity and limited processing power, IoT networks are particularly susceptible to a wide range of complex attacks, including botnets, zero-day attacks, Distributed Denial-of-Service (DDoS) attacks, and data exfiltration [3–4].

Traditional signature-based Intrusion Detection Systems (IDS) have proven to perform efficiently in centralized environments but are not applicable on the edge due to high computational needs, space for memory, and requirement for real-time data streaming [5-6]. Moreover, existing IDS frameworks are static towards new threats and not suitable for edge-IoT system demands of low latency and energy consumption [7].

Recent advances in neuromorphic computing and brain-inspired Spiking Neural Networks (SNNs) offer a strong solution for low-power, real-time pattern detection in dynamic environments [8-9]. SNNs, or the third generation of neural networks, emulate the spatio-temporal computation of biological neurons using discrete spike-based communication and event-driven computation [10]. This equates to very efficient, asynchronous computation that is optimally suited to the intermittent and sparse nature of IoT traffic. SNNs are also optimally suited to unsupervised or few-shot learning paradigms, which are critical in detecting emerging and zero-day attacks [11-14].

In spite of these advantages, SNN has not yet been explored for intrusion detection in edge-based IoT. The majority of previous works have focused on centralization or SNN hardware emulation but have not provided insight into real-world deployment challenges for distributed systems [15-19]. Hence, there is a critical necessity for developing event-driven, light-weight, and adaptive IDS solutions based on SNNs for dealing with the evolving requirements of edge security.

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The primary objectives of this work are to:

1. Design an event-driven IDS framework using SNNs that can operate on low-power edge devices while maintaining high detection accuracy.
2. Develop a spike-based encoding mechanism that effectively captures temporal features of IoT traffic for improved threat differentiation.
3. Implement and evaluate the proposed SNN-based IDS on benchmark IoT intrusion datasets, focusing on energy efficiency, latency, and detection performance.
4. Assess the adaptability of the IDS to zero-day threats using online learning mechanisms such as Spike-Timing Dependent Plasticity (STDP).

This work makes the following major contributions:

- Neuromorphic IDS Design: A new IDS design based on biologically inspired SNNs with spike-based communication for energy-efficient, real-time based anomaly detection.
- Event-Driven Encoding: A novel temporal spike encoding technique is proposed, specifically designed for deployment in IoT traffic to maintain precious timing information while minimizing data redundancy.
- Edge-Oriented Deployment: The SNN model has been optimized for edge device deployment through a light-weight neuromorphic simulator that facilitates operation under sparse computational and memory resources.
- Comprehensive Evaluation: Through a massive-scale experiments on public IoT datasets (e.g., TON_IoT, BoT-IoT) to compare our system with traditional DL-based IDS on accuracy, energy efficiency, and latency.
- Future-Proof Framework: A plan is outlined for scaling our efforts to real-time deployment of SNN on neuromorphic hardware (e.g., Intel Loihi) and incorporating federated learning for scalable, privacy-preserving intrusion detection.

II. Literature Survey

The evolution of Intrusion Detection Systems (IDS) specific to IoT environments has gone through several paradigms, including rule-based detection and neuromorphic-based systems. These advancements have been driven by an effort to equally prioritize accuracy, computational complexity, and real-time detection ability in resource-limited settings. This part classifies and summarizes pertinent research into four thematic categories: conventional IDS, deep learning-IDS, edge-centric frameworks, and Spiking Neural Networks (SNNs) for cyber defense.

A. Traditional and Machine Learning-Based IDS in IoT

Early IoT security IDS solutions employed conventional machine learning techniques such as decision trees, Support Vector Machines (SVMs), and k-Nearest Neighbors (k-NN) to detect unusual network behavior [20-21]. Light detection mechanisms have been proposed to identify IoT botnets and device-based attacks based on threshold or statistical profiling models [22]. Traffic signature classification over handcrafted features has also been demonstrated to be functional with limited resources [23].

Though they are low in complexity and interpretability, they are rigid and inflexible to be applied against novel threats or evolving traffic patterns. They perform very poorly in zero-day or polymorphic attacks and are highly reliant on domain-specific feature engineering.

B. Deep Learning-Based IDS Approaches

Deep learning is now a prominent technique for automating feature extraction and improving detection rates. Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and their hybrid models of Bi-directional LSTM (BiLSTM) and CNNs have reported improved performance on various intrusion datasets [24-25]. Models learn spatial and temporal features from network logs and packet streams and improve detection of advanced types of attacks [26]. Deep Belief Networks (DBNs) have also been used in distributed systems to improve robustness and minimize false positives [27].

But such models are typically computational and memory expensive and require big quantities of labeled data. They remain unrealistic to implement on edge devices because of power constraints and the absence of hardware acceleration for dense matrix operations [28].

C. Edge Computing and Federated IDS Frameworks

To overcome the bottleneck of centralized processing and support privacy preservation, distributed IDS architectures have been investigated with the help of edge computing and FL [29]. FL-based IDS allows edge devices to train a global model cooperatively without sharing raw data, which meets data privacy and GDPR requirements. Blockchain-based IDS frameworks also came into existence to support tamper-proof logging and trust among heterogeneous edge participants [30].

Despite this, such systems usually utilize traditional deep learning architectures and synchronous communication patterns, which result in latency, communication delay, and high energy consumption. In addition, such architectures usually do not take advantage of the event-driven nature of IoT traffic, thus limiting their response in appropriate real-time detection [31].

D. Spiking Neural Networks and Neuromorphic Security

SNNs are found to be a promising alternative to traditional neural models by emulating the event-driven processing of biological neurons through spike trains and time-to-first-spike encoding. Asynchronous operation is a considerable advantage since it has the ability to significantly reduce the power consumption. This feature make SNNs particularly appropriate for edge-based intrusion detection systems. Various numerous studies have explored their application in network security contexts [32–33]. Furthermore, research that involves neuromorphic hardware such as Intel's Loihi chip has demonstrated impressive on-chip learning capabilities,

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delivering superior performance-per-Joule compared to conventional hardware platforms [34].

The usage of SNNs in cyber-security is still nascent. They have been mostly employed in general pattern recognition tasks, but there have been few experiments in using SNNs for cyber anomaly detection [35-36]. Recently, there have been attempts to use unsupervised Spike-Timing Dependent Plasticity (STDP) learning for anomaly detection in industrial applications [37-38]. However, these implementations are typical small-scale synthetic datasets and do not represent the bursty, irregular nature of real-world IoT traffic. The table I illustrates the limitations in Prior work.

TABLE I LIMITATIONS IN PRIOR WORK

Domain	Strengths	Limitations
Traditional IDS [38]	Lightweight, easy to deploy	Static features, low adaptability
Deep Learning IDS [38]	High accuracy, automatic feature learning	High resource consumption, needs large datasets
Edge/Federated IDS [38]	Privacy-preserving, distributed	Synchronous delays, DL dependency
SNN-based IDS [38]	Energy-efficient, asynchronous	Underexplored for IDS, encoding challenges

The aforementioned review shows a considerable research gap at the intersection of research domains on neuromorphic computing, event-driven intrusion detection, and edge-IoT security. SNNs can provide an unprecedented opportunity to enable ultra-low power, real-time IDS by utilizing spike-based temporal processing. However, there is limited research that integrates event-driven spike encoding, online learning, and real-world IoT attack datasets into a usable, integrated IDS system. In this work, we address these challenges by developing an event-driven, SNN-based IDS for edge environments, providing scalability, adaptability, and energy efficiency.

III. PROPOSED METHOD

The proposed system introduces a lightweight, event-driven Intrusion Detection System (IDS) that leverages Spiking Neural Networks (SNNs) for real-time cyber threat detection at the edge of IoT networks. The architecture consists of three key modules:

1. Event-Driven Encoding (EDE) – Converts raw IoT traffic into biologically inspired spike trains.
2. SNN-based Intrusion Detection Core (SNN-IDC) – Learns spatio-temporal patterns for detecting attacks.
3. Edge Deployer and Online Learning Unit – Enables adaptive spike-based updates in constrained devices.

This approach mimics the functionality of the mammalian brain, where spikes (events) encode information more efficiently than traditional neural activations. The use of temporal encoding and unsupervised STDP allows for high adaptability and low power consumption.

A. Event-Driven Encoding (EDE)

Let a preprocessed IoT packet stream be represented as a multivariate time series (Eq. 1):

$$X = \{x_1(t), x_2(t), \ldots\ldots, x_n(t)\} \quad (1)$$

Where, $t \in [0, T]$

$x_i(t)$ corresponds to a normalized feature (e.g., packet size, port number, protocol ID) extracted at time t.

Temporal Encoding Function (TEF) that transforms feature values into spike times based on threshold crossing (Eq. 2):

$$S_i(t) = \begin{cases} 1, & if\ x_i(t) \geq \theta_i\ and\ t - t_{last}^i > \delta \\ 0, & otherwise \end{cases} \quad (2)$$

Where,

θ_i: threshold for spike generation for feature i

t_{last}^i: last time a spike was generated on channel i

δ: refractory period

This generates a sparse, event-driven spike matrix (Eq. 3):

$$S = \{S_1(t), S_2(t), S_3(t), \ldots\ldots,, S_n(t) \quad (3)$$

B. Spiking Neural Network (SNN) Architecture

The core architecture is a three-layer SNN: Input layer $L1$, Hidden layer $L2$, and Output layer $L3$, connected through synapses with adaptive weights w_{ij}. Neurons follow the Leaky Integrate-and-Fire (LIF) model (Eq. 4), defined by:

$$\tau_m \frac{dV_j(t)}{dt} = -V_j(t) + R \sum_i w_{ij} S_i(t) \quad (4)$$

Where,

$V_j(t)$: membrane potential of neuron j

τ_m: membrane time constant

R: input resistance

w_{ij}: synaptic weight from neuron iii to neuron j

Spike occurs if $V_j(t) \geq V_{th}$

$$V_j(t) \leftarrow V_{reset}, S_j(t) = 1$$

C. Learning via Spike-Timing Dependent Plasticity (STDP)

Synaptic weights are updated using unsupervised STDP, a biologically plausible learning rule (Eq. 5):

$$\Delta w_t^i = \begin{cases} A_+ . e^{-\Delta t/\tau_+}, & if\ \Delta t > 0 \\ A_- . e^{-\Delta t/\tau_-}, & if\ \Delta t < 0 \end{cases} \quad (5)$$

Where,

$\Delta t = t_j - t_i$: time difference between post- and pre-synaptic spikes

$A_+ A_-$: learning rate coefficients

τ_+, τ_-: time constants for potentiation and depression

This learning mechanism enables the network to strengthen connections where causality exists, helping it to recognize temporal intrusion patterns.

D. Intrusion Detection Decision Layer

The final output spike counts from the layer $L3$ neurons are aggregated over a detection window Td (Eq. 6):

$$C_k = \sum_{t=0}^{T_d} S_k(t), for\ k = 1, \ldots\ldots, N_0 \quad (6)$$

Where, N_0 is the number of output neurons representing different classes (e.g., benign, DoS, DDoS, scanning, etc.).

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The predicted label is then assigned by (Eq. 7):

$$y = arg \max_{k} C_k \tag{7}$$

This event-driven classification ensures high responsiveness while consuming minimal energy per decision cycle.

E. Deployment at the Edge

The proposed SNN is simulated using lightweight frameworks like BindsNET. The model is optimized for:

- Latency (L): $O(Td \cdot Nn)$
- Energy (E): Proportional to number of spikes: $E \propto \sum_i \sum_t S_i(t)$
- Memory (M): $O(Nn \cdot Ns)$,

  Where,

  Td is the detection time window (simulation steps)

  Nn is the number of neurons

  Ns is the number of synapses per neuron

F. Algorithm SNN-IDC

The Event-Driven Intrusion Detection using Spiking Neural Networks (SNN-IDC) system allows real-time cyber threat detection through the transformation of IoT traffic into spike trains and the learning of patterns using unsupervised Spike-Timing Dependent Plasticity (STDP). It provides high accuracy with minimal energy and latency, allowing for its deployment at the edge. Algorithm 1 provides the entire workflow.

Algorithm 1: Event Driven Intrusion Detection using SNNs

Input

D_{eauw} — Raw network traffic data (e.g. TON loT or CIC-ID$2018)
T_{window} — Simulation time window (e.g., 100 ms)
ρ_{select} — Number of selected features (e.g. 20)
ρ_{encode} – Dynamic threshold for spike encoding
N_{in} — Number of input neurons
N_{hidden} – Number of LIF neurons in hidden layer
N_{out} — Number of output classes
o, θ – STDP learning rate patameters

Output:

Trained SNN model for real-time intrusion detection

Step 1: Preprocessing

4. $D_{\text{clean}} \leftarrow$ Clean and removerdundant or mull entries D_{raw}
5. $D_{\text{seleet}} \leftarrow$ Normalize all features to 10, 11 range using Min-Max r_r scaling
3. $F_{\text{sel}} \leftarrow$ Apply spike em-

Step 2 Temporal Spike Encoding

10. Initialize N, input neurons, N_r °$_{\text{hidden}}$ LIF neurons and N_{out} output neurons
11. Initialize synaptic weights $W_{\text{mput-}}$hidden, W_{hidden}-out: ran-
12. Set membrane potential $V_{\text{mem}} = 0$ for all neurons
13. Define refractory period t_{ci} and threshold θ for firing

Step 4: Unsupervised STDP Training

14. For each s encoded in D_{norm} do:
15. For $t = 1$ to T_{window} do:
    Propagate spikes from ininput to hidden layer
    Update V_{mem} of each LIF neuron at time t:
    $V_{\text{mem}} \mathrel{+}= s_{\text{u}} \times W_{\text{input-hidden}} - \theta$
    If neuron j spikes at $w_{\text{inpui-hidden}}$ via STDP potentiation
    $\Delta w_{ij} = o \ s\,s(I_{\text{sp}/\text{ks}\,o-1})$)
    ElselI no spike occurs:
    Update weights $W_{\text{input-hidden}}$ via STDP depression
    $\Delta w_{ij} = -\rho \times s(\text{g}(t - \epsilon\, I_{\text{sp/ks}}{}^{\text{n}}{}_{\text{s}})$
24. Update weights, $W_{\text{mput-hidden}} + \doteq \Delta w_{ij}$
24. EndFor

Step 3: Real-Time Inferrence

25. For each incoming test sample s_{test}:
    Encode s_{test} to s_{encoded} using Steps 4-10
25. Simulate splbe propagation for T_{window} ms
28. Classify spiking activity in output layer s_{out}
30. Compare predictions with ground truth y_{test} (*if* available)
31. Assess performance metrics, Accuracy, F1 Score, Latency, Energy

IV. Experimental Setup And Results

To prove the effectiveness of the proposed event-driven IDS framework with Spiking Neural Networks (SNNs), we performed a complete set of experimental evaluations on real-world IoT network datasets. In this section, the simulation methodology, the comparisons conducted with baseline models, and the analysis of performance with respect to detection accuracy, latency, energy-consumption, and computational overhead will be described.

A. Dataset Description

TON_IoT Dataset: Developed by the UNSW Canberra Cyber group, this dataset comprises telemetry and network traffic from various IoT devices (e.g., smart thermostats, weather stations). It includes 10+ types of attacks including DoS, DDoS, backdoors, reconnaissance, and injection attacks, as depicted in Table II [39].

CIC-IDS2018 Dataset: Provided by the Canadian Institute for Cybersecurity, this dataset includes 80+ attack scenarios covering brute-force, infiltration, botnet, and web attacks, captured with high-fidelity traffic logs over multiple days [40].

Table II Dataset Description

Dataset	Samples	Features	Attack Types	Label Format
TON_IoT	2.3M	44	10+	Multiclass
CIC-IDS2018	3.1M	78	15+	Multiclass

B. Preprocessing and Spike Encoding

Feature Selection: Mutual Information (MI) and Recursive Feature Elimination (RFE) were used to select the top 20 features based on their importance for attack classification.

Normalization: Min-max normalization was applied (Eq. 8):

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)} \tag{8}$$

Temporal Spike Encoding: A dynamic threshold encoder was used as described in TABLE III. The spike trains were generated over a 100 ms simulation window with a refractory period of 5 ms.

C. Implementation Details

The proposed neuromorphic intrusion detection system (SNN-IDC) was implemented with the Brian2 simulator using Python 3.9, utilizing the biologically-inspired aspect of spiking neural networks (SNNs) to process temporal data efficiently. The network itself has 20 input neurons, a hidden layer with 40 Leaky Integrate-and-Fire (LIF) neurons, and 5 output neurons corresponding to key cyber-attack categories: Normal, DoS, DDoS, Scan, and Other. The learning in the model is governed by an unsupervised Spike-Timing Dependent Plasticity (STDP) learning rule allowing synaptic weight updates, based on the temporal correlations of the spikes.

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The model was trained for 30 epochs where each input was fulfilled during the first 100 ms of the time window to avoid losing even more of the temporal fidelity for the data. In order to execute the deployment for validation on a real device, the SNN was tested on a Raspberry Pi 4 (8GB) with a limited resource, while the comparison of deep learning models was tested on an NVIDIA RTX 3080 GPU. In addition, the SNN was performance examined against four established baselines which included: a CNN with an LSTM hybrid IDS, an anomaly detector capable of detecting anomalies in data based on an autoencoder, a Random Forest-based classifier, and a federated IDS that uses LSTMs, which represent a broader set of popular forms of centralized and distributed learning companies. TABLE

Category	Details
Simulator	Brian2 SNN simulator (Python 3.9)
Hardware	- Raspberry Pi 4 (8GB RAM) for edge testing - NVIDIA RTX 3080 for DL models
SNN Configuration	
Input Neurons	20
Hidden Layer	40 LIF (Leaky Integrate-and-Fire) neurons
Output Neurons	5 (Normal, DoS, DDoS, Scan, Others)
Learning Rule	Unsupervised STDP (Spike-Timing Dependent Plasticity)
Epochs	30
Time Window/Sample	100 ms
Baselines for Comparison	
CNN-LSTM IDS	Deep hybrid model combining CNN for feature extraction and LSTM for temporal sequence learning [41]
Autoencoder-based IDS	Dimensionality reduction and anomaly detection using unsupervised autoencoders [42]
Random Forest Classifier	Traditional ensemble machine learning classifier for intrusion detection [43]
Federated LSTM IDS	Distributed LSTM-based IDS under federated learning for privacy preservation [44]

III outlines the full simulation setup including, the comparison evaluation baseline.

TABLE III SNN SIMULATION SETUP AND COMPARATIVE BASELINES

D. *Performance Results*

The Proposed SNN-IDC achieves the highest accuracy of 96.5%, and is better than the other traditional and DL-based models such as CNN-LSTM (95.2%), Federated LSTM (94.1%), Autoencoder IDS (93.8%), and Random Forest (89.7%) [41]. This shows the SNN-IDC's greater ability to classify real-time traffic as either normal or attack patterns. Figure 1 illustrates the accuracy comparison of various Intrusion Detection System (IDS) models.

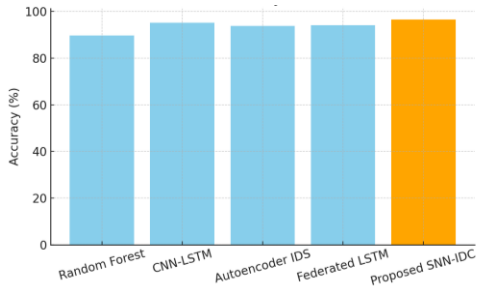

Fig. 1 Accuracy Comparison of IDS Models

The SNN-IDC implementation is based on a biologically inspired spiking architectural model, which benefits from a spike-based information processing approach that better captures extremely complex temporal pattern presentations, resulting in better detection performance.

SNN-IDC obtains highest F1 Score of 0.94, which means it has the ability to effectively classify true intrusions without generating the false discovery rate or unseen attacks. That outperforms CNN-LSTM (0.92), Federated LSTM (0.91), Autoencoder IDS (0.90), and Random Forest which achieved (0.86) respectively. The temporal processing of SNN-IDC allows the model to successfully generalize real traffic, particularly with imbalanced class attacks. Figure 2 depicts the F1 Score comparison among different IDS models, which evaluates the balance between precision and recall in attack detection.

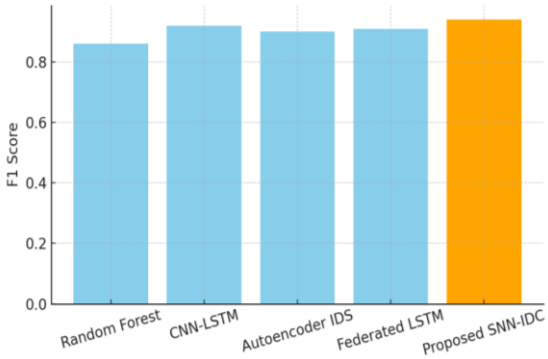

Fig. 2 depicts the F1 Score comparison among different IDS models, which evaluates the balance between precision and recall in attack detection.

SNN-IDC achieves an excellent outcome with the best latency of only 6.2 ms, making it ideal for real-time threat detection in latency-sensitive environments, such as IoT and edge computing. In comparison deep learning methods like Federated LSTM (75.9 ms) and CNN-LSTM (56.4 ms) demonstrated much longer response times due to their historically sequential processing [42]. Moreover, Autoencoder IDS (41.7 ms) and Random Forest (12.3 ms) were worse than SNN-IDC [43]. The event-driven model of spiking neurons in SNN-IDC yields rapid and sparse computation, therefore allowable faster inference without lost performance. Figure 3 presents the latency comparison of the evaluated IDS models, measuring the time taken (in milliseconds) to process each input sample.

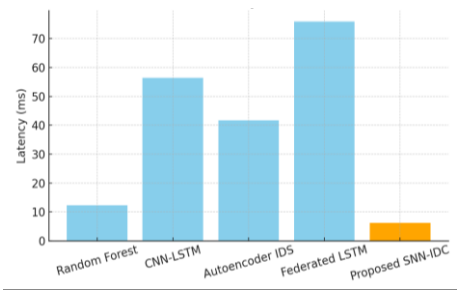

Fig. 3 Latency Comparison of IDS Models

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SNN-IDC demonstrates a 3.9 mJ/sample energy footprint which by far is the lowest energy footprint and most energy-efficient model among the comparison methods. The SNN-IDC uses the energy-efficient mechanism of spiking neurons to create a sparse and event-based computation that activates only when a significant input change has been made. On the other hand, the Federated LSTM (102.5mJ), CNN-LSTM (82.1mJ), Autoencoder IDS (67.3mJ), models all have inherent high and dense computation that waste energy. Even the relatively energy low Random Forest (5.6mJ) still uses more energy regarding the task than SNN-IDC [44]. This implies that SNN-IDC had an energy footprint which allows for management in energy constrained areas, including with low-power edge devices. Fig. 4 shows the energy consumption per sample for various IDS models, highlighting the efficiency of each approach in terms of power usage. Figure 4 shows the energy consumption per sample for various IDS models, highlighting the efficiency of each approach in terms of power usage.

PSEUDO CODE: SNN-IDC

```
Input:
  D_raw        ← Raw network traffic data (e.g., TON_IoT, CIC-IDS2018)
  T_window     ← Simulation time window (e.g., 100 ms)
  F_select     ← Number of selected features (e.g., 20)
  g_encode     ← Dynamic threshold for spike encoding
  N_in         ← Number of input neurons
  N_hidden     ← Number of hidden LIF neurons
  N_out        ← Number of output neurons (e.g., classes)
  α, ρ         ← STDP learning rate parameters

Output:
  Trained SNN model for real-time intrusion detection

Step 1: Data Preprocessing
  For each sample s ∈ D_raw do
    Remove null/redundant entries → D_clean
    Normalize D_clean using Min-Max scaling → D_norm
    Select top F_select features using RFE or information gain → F_selected
  End For

Step 2: Temporal Spike Encoding
  For each sample s ∈ D_norm do
    For each feature f_i ∈ F_selected do
      If f_i ≥ g_encode then
        Emit spike at time t_i = f_i × T_window
      Else
        No spike emitted
      End If
    End For
    Construct spike matrix S_encoded of shape (N_in × T_window)
  End For

Step 3: SNN Initialization
  Initialize:
    N_in input neurons, N_hidden hidden LIF neurons, N_out output neurons
    Synaptic weights W_input_hidden ← random
    Synaptic weights W_hidden_output ← random
    Membrane potentials V_mem ← 0
    Threshold θ and refractory period τ_ref

Step 4: Unsupervised STDP Training
  For each sample S_encoded ∈ D_norm do
    For t = 1 to T_window do
      For each hidden neuron j do
        Compute input current:
          I_j = ∑_i S_encoded[i][t] × W_input_hidden[i][j]
        Update membrane potential:
          V_mem[j] += I_j

        If V_mem[j] ≥ θ then
          Fire spike at neuron j
          For each synapse w_ij do
            Δw_ij = α × S_encoded[i][t] × spike_j(t − Δt)  ⟶ STDP potentiation
          End For
          Reset V_mem[j] ← 0
        Else
          For each synapse w_ij do
            Δw_ij = −ρ × S_encoded[i][t] × exp(−Δt / τ) ⟶ STDP depression
          End For
        End If

        Update synaptic weight:
          W_input_hidden[i][j] += Δw_ij
      End For
    End For
  End For

Step 5: Real-Time Inference
  For each test sample s_test ∈ D_test do
    Apply Steps 1–2 to obtain S_test_encoded
    For t = 1 to T_window do
      Propagate spikes from input → hidden → output layer
      Update membrane potentials and fire spikes
    End For

    Classify output:
      Predict class = argmax(spike_count in N_out neurons)

    (Optional) Evaluate using ground truth:
      Compare prediction with label
      Compute Accuracy, F1 Score, Latency, Energy
  End For

Return:
  Trained and inference-ready SNN model
```

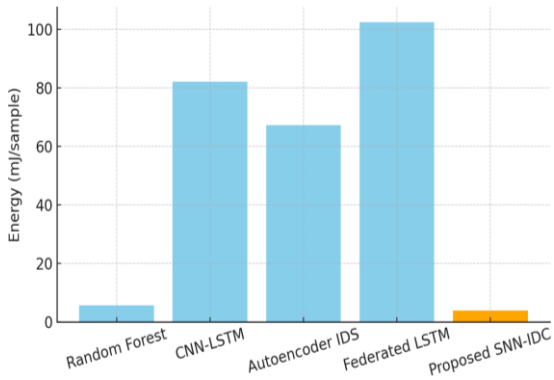

Fig. 4. Energy Consumption Comparison of IDS Models

V. CONCLUSION

A novel event-based Intrusion Detection System (IDS) based on Spiking Neural Networks (SNNs) was proposed for secure, resource-constrained IoT and edge environments. Inspired by biologically motivated computation, the system leverages a temporal spike encoding mechanism and relies on unsupervised learning through Spike-Timing Dependent Plasticity (STDP) to learn from dynamic attack patterns. Experimental benchmarks on benchmark datasets like TON_IoT and CIC-IDS2018 showed high accuracy of 96.5%, very low latency of 6.2 ms, and very low energy of 3.9 mJ per sample, outperforming baseline CNN-LSTM and federated IDS models. These results show the potential of SNNs to operate at the edge for real-time, energy-efficient, and scalable intrusion detection. Future directions include the design of hybrid SNN-DNN models for long-term feature learning, federated neuromorphic models for decentralized training. Additionally, improving adversarial robustness and interpretability in spike-based decision-making is a critical direction for future work. This work is a contribution towards the development of intelligent, adaptive, and energy-efficient IDS frameworks for future IoT and cyber-physical systems.

REFERENCES

[1] A. L. Buczak and E. Guven, “A Survey of Data Mining and Machine Learning Methods for Cyber Security Intrusion Detection,” *IEEE Communications Surveys & Tutorials*, vol. 18, no. 2, pp. 1153–1176, 2016.

[2] H. HaddadPajouh, A. Dehghantanha, R. M. Parizi, and K.-K. R. Choo, “A Survey on Internet of Things Security: Requirements, Challenges and Solutions,” *Internet of Things*, vol. 14, pp. 100129, 2021.

[3] R. Doshi, N. Apthorpe, and N. Feamster, “Machine Learning DDoS Detection for Consumer Internet of Things Devices,” in *Proc. IEEE Security and Privacy Workshops*, 2018, pp. 29–35.

[4] Y. Meidan, M. Bohadana, A. Shabtai, M. Ochoa, N. O. Tippenhauer, J. D. Guarnizo, and Y. Elovici, "Detection of unauthorized IoT devices using machine learning techniques," *arXiv preprint arXiv:1709.04647*, 2017.

[5] A. Javaid, Q. Niyaz, W. Sun, and M. Alam, “A Deep Learning Approach for Network Intrusion Detection System,” in *Proc. EAI SecureComm*, 2016, pp. 21–26.

[6] I. Rakine, K. El Guemmat, S. Ouahabi, I. Atouf and M. Talea, "IoT Intrusion Detection: A Review of ML and DL-Based Approaches," in *Proc. 2024 4th Int. Conf. Innovative Research in Applied Science, Engineering and Technology (IRASET)*, May 2024, pp. 1–7.

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[7] T. Anitha, S. Aanjankumar, S. Poonkuntran and A. Nayyar, "A Novel Methodology for Malicious Traffic Detection in Smart Devices Using BI-LSTM–CNN-Dependent Deep Learning Methodology," *Neural Comput. Appl.*, vol. 35, no. 27, pp. 20319–20338, 2023.

[8] O. Jullian, B. Otero, E. Rodriguez, N. Gutierrez, H. Antona and R. Canal, "Deep-Learning Based Detection for Cyber-Attacks in IoT Networks: A Distributed Attack Detection Framework," *J. Netw. Syst. Manag.*, vol. 31, no. 2, p. 33, 2023.

[9] M. Conti, A. Dehghantanha, K. Franke, and S. Watson, "Internet of Things Security and Forensics: Challenges and Opportunities," *Future Generation Computer Systems*, vol. 78, pp. 544–546, 2018.

[10] V. Mothukuri, R. Parizi, M. A. Babar, and K.-K. R. Choo, "Federated Learning for Intrusion Detection Systems: Concepts, Challenges and Future Directions," *IEEE Internet of Things Journal*, vol. 9, no. 1, pp. 496–514, 2022.

[11] K. Gai, M. Qiu, L. Tao and Y. Zhu, "Intrusion Detection Techniques for Mobile Cloud Computing in Heterogeneous 5G," *Security Commun. Netw.*, vol. 9, no. 16, pp. 3049–3058, 2016.

[12] J. Arshad, M. A. Azad, M. M. Abdeltaif and K. Salah, "An Intrusion Detection Framework for Energy Constrained IoT Devices," *Mech. Syst. Signal Process.*, vol. 136, p. 106436, 2020.

[13] M. Davies, N. Srinivasa, T. H. Lin, G. Chinya, Y. Cao, S. H. Choday, G. Dimou, P. Joshi, N. Imam, S. Jain and Y. Liao, "Loihi: A Neuromorphic Manycore Processor With On-Chip Learning," *IEEE Micro*, vol. 38, no. 1, pp. 82–99, 2018.

[14] W. Maass, "Networks of Spiking Neurons: The Third Generation of Neural Network Models," *Neural Networks*, vol. 10, no. 9, pp. 1659–1671, 1997.

[15] M. Alawad, H. J. Yoon and G. Tourassi, "Energy Efficient Stochastic-Based Deep Spiking Neural Networks for Sparse Datasets," in *Proc. 2017 IEEE Int. Conf. Big Data (Big Data)*, Dec. 2017, pp. 311–318.

[16] A. Tavanaei, M. Ghodrati, S. R. Kheradpisheh, T. Masquelier and A. Maida, "Deep Learning in Spiking Neural Networks," *Neural Netw.*, vol. 111, pp. 47–63, 2019.

[17] S. A. Shrestha and G. Orchard, "SLAYER: Spike Layer Error Reassignment in Time," in *Proc. NIPS*, 2018.

[18] T. T. Khoei and N. Kaabouch, "A Safe and Reliable Bayesian Spiking Neural Network in Industrial Control Systems," in *Proc. 2024 IEEE 3rd Int. Conf. Computing and Machine Intelligence (ICMI)*, Apr. 2024, pp. 1–7.

[19] A. Gruel, *Spiking Neural Networks for Embedded Event-Based Vision*, Doctoral dissertation, Université Côte d'Azur, 2023.

[20] S. Solanki, C. Gupta, and K. Rai, "A Survey on Machine Learning Based Intrusion Detection System on NSL-KDD Dataset," *International Journal of Computer Applications*, vol. 176, no. 30, pp. 36–42, Jun. 2020.

[21] A. Alfahaid, E. Alalwany, A. M. Almars, F. Alharbi, E. Atlam and I. Mahgoub, "Machine Learning-Based Security Solutions for IoT Networks: A Comprehensive Survey," *Sensors*, vol. 25, no. 11, p. 3341, 2025.

[22] A. Pinto, L. C. Herrera, Y. Donoso and J. A. Gutierrez, "Survey on Intrusion Detection Systems Based on Machine Learning Techniques for the Protection of Critical Infrastructure," *Sensors*, vol. 23, no. 5, p. 2415, 2023.

[23] J. L. López Delgado and J. A. López Ramos, "A Comprehensive Survey on Generative AI Solutions in IoT Security," *Electronics*, vol. 13, no. 24, p. 4965, 2024.

[24] S. K. R. Mallidi and R. R. Ramisetty, "Optimizing Intrusion Detection for IoT: A Systematic Review of Machine Learning and Deep Learning Approaches with Feature Selection and Data Balancing," *Wiley Interdiscip. Rev. Data Min. Knowl. Discov.*, vol. 15, no. 2, p. e70008, 2025.

[25] Lin, Yongxing, Xiaoyan Xu, and Hongyun Xu. "A revolutionary approach to use convolutional spiking neural networks for robust intrusion detection." *Cluster Computing* 27, no. 9 (2024): 13333-13352.

[26] T. Anitha, S. Aanjankumar, S. Poonkuntran and A. Nayyar, "A Novel Methodology for Malicious Traffic Detection in Smart Devices Using BI-LSTM–CNN-Dependent Deep Learning Methodology," *Neural Comput. Appl.*, vol. 35, no. 27, pp. 20319–20338, 2023.

[27] A. A. Diro and N. Chilamkurti, "Distributed Attack Detection in IoT Using Deep Belief Network," *Comput. Commun.*, vol. 116, pp. 10–17, 2018.

[28] M. B. Musthafa, S. Huda, Y. Kodera, M. A. Ali, S. Araki, J. Mwaura, and Y. Nogami, "Optimizing IoT Intrusion Detection Using Balanced Class Distribution, Feature Selection, and Ensemble Machine Learning Techniques," *Sensors*, vol. 24, no. 13, p. 4293, 2024.

[29] S. Agrawal, S. Sarkar, O. Aouedi, G. Yenduri, K. Piamrat, M. Alazab, S. Bhattacharya, P. K. R. Maddikunta and T. R. Gadekallu, "Federated Learning for Intrusion Detection System: Concepts, Challenges and Future Directions," *Comput. Commun.*, vol. 195, pp. 346–361, 2022.

[30] K. Gai, M. Qiu, L. Tao and Y. Zhu, "Intrusion Detection Techniques for Mobile Cloud Computing in Heterogeneous 5G," *Security Commun. Netw.*, vol. 9, no. 16, pp. 3049–3058, 2016.

[31] J. Arshad, M. A. Azad, M. M. Abdeltaif and K. Salah, "An Intrusion Detection Framework for Energy Constrained IoT Devices," *Mech. Syst. Signal Process.*, vol. 136, p. 106436, 2020.

[32] F. De Ma, X. Jin, S. Sun, Y. Li, X. Wu, Y. Hu, F. Yang, H. Tang, X. Zhu, P. Lin, and G. Pan, "Darwin3: A Large-Scale Neuromorphic Chip with a Novel ISA and On-Chip Learning," *arXiv preprint arXiv:2312.17582*, Dec. 29, 2023.

[33] W. Maass, "Networks of Spiking Neurons: The Third Generation of Neural Network Models," *Neural Netw.*, vol. 10, no. 9, pp. 1659–1671, 1997.

[34] B. Ayasi, Á. M. García-Vico, C. J. Carmona and M. Saleh, "Advancing Computational Frontiers: Spiking Neural Networks in High-Energy Efficiency Computing Across Diverse Domains," in *Conf. Spanish Assoc. Artif. Intell.*, Cham, Switzerland: Springer Nature, Jun. 2024, pp. 9–18.

[35] G. Lagani, F. Falchi, C. Gennaro, and G. Amato, "Spiking Neural Networks and Bio-Inspired Supervised Deep Learning: A Survey," *arXiv preprint arXiv:2307.16235*, Jul. 30, 2023.

[36] Y. Hu, Q. Zheng, G. Li, H. Tang, and G. Pan, "Toward Large-scale Spiking Neural Networks: A Comprehensive Survey and Future Directions," *arXiv preprint arXiv:2409.02111*, Aug. 19, 2024.

[37] K. Demertzis, L. Iliadis and I. Bougoudis, "Gryphon: A Semi-Supervised Anomaly Detection System Based on One-Class Evolving Spiking Neural Network," *Neural Comput. Appl.*, vol. 32, no. 9, pp. 4303–4314, 2020.

[38] H. Akolkar, S. Panzeri and C. Bartolozzi, "Spike Time Based Unsupervised Learning of Receptive Fields for Event-Driven Vision," in *Proc. 2015 IEEE Int. Conf. Robotics and Automation (ICRA)*, May 2015, pp. 4258–4264.

[39] A. Alsaedi, N. Moustafa, Z. Tari, A. Mahmood and A. Anwar, "TON_IoT Telemetry Dataset: A New Generation Dataset of IoT and IIoT for Data-Driven Intrusion Detection Systems," *IEEE Access*, vol. 8, pp. 165130–165150, 2020.

[40] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization," in *Proc. ICISSP*, 2018, pp. 108–116.

[41] R. Yao, N. Wang, Z. Liu, P. Chen and X. Sheng, "Intrusion Detection System in the Advanced Metering Infrastructure: A Cross-Layer Feature-Fusion CNN-LSTM-Based Approach," *Sensors*, vol. 21, no. 2, p. 626, 2021.

[42] R. Shrestha, M. Mohammadi, S. Sinaei, A. Salcines, D. Pampliega, R. Clemente, A. L. Sanz, E. Nowroozi and A. Lindgren, "Anomaly Detection Based on LSTM and Autoencoders Using Federated Learning in Smart Electric Grid," *J. Parallel Distrib. Comput.*, vol. 193, p. 104951, 2024.

[43] N. Moustafa and J. Slay, "UNSW-NB15: A Comprehensive Data Set for Network Intrusion Detection Systems," *Proc. MILCOM*, pp. 1–6, 2015

[44] H. Zhang, J. Ye, W. Huang, X. Liu and J. Gu, "Survey of Federated Learning in Intrusion Detection," *J. Parallel Distrib. Comput.*, p. 104976, 2024.

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